Magnesium sulfate

Neonatal

FORM Ampoule containing magnesium sulfate 50%
2 mmol per 1 ml ampoule

INDICATION Hypomagnesaemia, hypocalcaemia

DOSE RANGE

AGE	DOSE	FREQUENCY	ROUTE
Neonate	0.4 mmol/kg	6-12 hourly PRN	IV
1-6 months	0.2 mmol/kg	12 hourly PRN	IV

RECONSTITUTION Already in solution, **must be diluted prior to use**

DILUTION For peripheral administration use a dilution resulting in a 5% solution.

Concentrations above 5% have a high osmolarity and must be given via central access only. A large bore peripheral vein may be used in an emergency with caution. Maximum concentration 20% in fluid restriction.

	Peripheral 5%	Central only 10%	Central only 20%
Magnesium sulfate 50%	1 ml (2 mmol)	1 ml (2 mmol)	1 ml (2 mmol)
Sodium chloride 0.9%	9 ml	4 ml	1.5 ml
Final solution	2 mmol in 10 ml (= 0.2 mmol/ml)	2 mmol in 5 ml (= 0.4 mmol/ml)	2 mmol in 2.5 ml (= 0.8 mmol/ml)

When preparing magnesium sulfate mix the product thoroughly to avoid 'layering'

METHOD OF ADMINISTRATION Short IV infusion over at least 10 minutes (i.e. max 0.04mmol/kg/minute)

COMPATIBILITY

Solution compatibility	Sodium chloride 0.45%, sodium chloride 0.9%, glucose 5%, glucose 10%
Solution incompatibility	No information
IV Line compatibility	Aciclovir, Adrenaline, calcium gluconate, cefotaxime, clonidine, dobutamine, dopamine, fentanyl, gentamicin, insulin, labetalol, meropenem, metronidazole, midazolam, milrinone, morphine, noradrenaline, piperacillin/tazobactam, sodium bicarbonate, vancomycin, vecuronium.
IV Line incompatibility	Amiodarone, AmBisome, calcium chloride, hydrcortisone

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS	See Summary of Product Characteristics or BNF for Children
FURTHER INFORMATION	Magnesum sulfate 1 g = Mg 4 mmol Monitor serum magnesium, calcium, blood pressure and heart rate, especially with repeated dosing Effect of neuromuscular blockade may be enhanced
LICENSED STATUS	Not licensed for monotherapy in neonates
APPLICABLE POLICIES	West of Scotland Neonatal Guidelines Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department